

Lumen — Portfolio Documentation

AI-Based Web Accessibility Checker

Repository: github.com/balisikh/ai-based-web-accessibility-checker · Local:
<http://localhost:4376>

Assistive findings only — not a legal accessibility certificate or formal conformance audit.

Overview

Lumen scans a public URL with Playwright + axe-core, returns a score, severities, rule help, optional AI tips, and JSON export.

1. Paste a public website URL
2. Headless Chrome renders the page
3. axe-core checks WCAG A/AA-oriented rules
4. Score, severities, and issue details
5. Optional AI tips; JSON export

UI screenshots

Batch results — 28 sites

Website Search Results									
Rank	URL	Score	Category	Page Title	Page Content	Page Length	Page Date	Page Status	Page Type
1	http://www.example.com	95	Business	Example Corp	Example Corp is a leading provider of business solutions.	1000	2023-01-01	Active	Business
2	http://www.example2.com	90	Technology	Example Tech	Example Tech offers innovative software solutions.	800	2023-01-02	Active	Technology
3	http://www.example3.com	85	Healthcare	Example Health	Example Health provides comprehensive healthcare services.	1200	2023-01-03	Active	Healthcare
4	http://www.example4.com	80	Education	Example Edu	Example Edu is a leading provider of educational services.	900	2023-01-04	Active	Education
5	http://www.example5.com	75	Finance	Example Fin	Example Fin offers a range of financial services.	1100	2023-01-05	Active	Finance
6	http://www.example6.com	70	Real Estate	Example Real Estate	Example Real Estate provides real estate services.	1300	2023-01-06	Active	Real Estate
7	http://www.example7.com	65	Travel	Example Travel	Example Travel offers travel services.	700	2023-01-07	Active	Travel
8	http://www.example8.com	60	Food & Beverage	Example Food	Example Food provides food and beverage services.	1400	2023-01-08	Active	Food & Beverage
9	http://www.example9.com	55	Automotive	Example Auto	Example Auto offers automotive services.	600	2023-01-09	Active	Automotive
10	http://www.example10.com	50	Energy	Example Energy	Example Energy provides energy services.	1500	2023-01-10	Active	Energy
11	http://www.example11.com	45	Media	Example Media	Example Media offers media services.	500	2023-01-11	Active	Media
12	http://www.example12.com	40	Telecommunications	Example Telecom	Example Telecom provides telecommunications services.	1600	2023-01-12	Active	Telecommunications
13	http://www.example13.com	35	Construction	Example Construction	Example Construction offers construction services.	400	2023-01-13	Active	Construction
14	http://www.example14.com	30	Retail	Example Retail	Example Retail provides retail services.	1700	2023-01-14	Active	Retail
15	http://www.example15.com	25	Healthcare	Example Health	Example Health provides healthcare services.	300	2023-01-15	Active	Healthcare
16	http://www.example16.com	20	Education	Example Edu	Example Edu provides educational services.	1800	2023-01-16	Active	Education
17	http://www.example17.com	15	Finance	Example Fin	Example Fin provides financial services.	200	2023-01-17	Active	Finance
18	http://www.example18.com	10	Real Estate	Example Real Estate	Example Real Estate provides real estate services.	1900	2023-01-18	Active	Real Estate
19	http://www.example19.com	5	Travel	Example Travel	Example Travel provides travel services.	100	2023-01-19	Active	Travel
20	http://www.example20.com	0	Food & Beverage	Example Food	Example Food provides food and beverage services.	2000	2023-01-20	Active	Food & Beverage
21	http://www.example21.com	95	Business	Example Corp	Example Corp is a leading provider of business solutions.	1000	2023-01-01	Active	Business
22	http://www.example22.com	90	Technology	Example Tech	Example Tech offers innovative software solutions.	800	2023-01-02	Active	Technology
23	http://www.example23.com	85	Healthcare	Example Health	Example Health provides comprehensive healthcare services.	1200	2023-01-03	Active	Healthcare
24	http://www.example24.com	80	Education	Example Edu	Example Edu is a leading provider of educational services.	900	2023-01-04	Active	Education
25	http://www.example25.com	75	Finance	Example Fin	Example Fin offers a range of financial services.	1100	2023-01-05	Active	Finance
26	http://www.example26.com	70	Real Estate	Example Real Estate	Example Real Estate provides real estate services.	1300	2023-01-06	Active	Real Estate
27	http://www.example27.com	65	Travel	Example Travel	Example Travel offers travel services.	700	2023-01-07	Active	Travel
28	http://www.example28.com	60	Food & Beverage	Example Food	Example Food provides food and beverage services.	1400	2023-01-08	Active	Food & Beverage

Results — sample layout

L

Lumen

Checker

Batch results

Light

Dark

System

← New scan

Lumen

Results

https://www.w3.org/WAI/demos/bad/before/home.html

42

SCORE

Needs work

BATCH: FAIL

Critical: 3. 3 critical issues; score 42 (need ≥ 85)

3 critical issues — address these before lower-severity items, even if the score looks acceptable.

Score 0–100 from automated issue counts. **Needs work** means 42 (85+ is Strong). In our website test batches, **Pass** also requires zero critical issues.

CRITICAL

3

SERIOUS

2

MODERATE

5

MINOR

1

CRITICAL

Blocks many users — fix first.

SERIOUS

Major barriers for some users.

Filter severity

All

Export JSON

New scan

Assistive findings only — not a legal accessibility certificate.

CRITICAL

Images must have alternate text: decorative banner missing alt

SERIOUS

Elements must meet minimum color contrast ratio thresholds

CRITICAL

Images must have alternate text

WCAG

1.1.1

Rule

image-alt

Selector

img.hero-banner

SNIPPET

Mobile home (390px)

Layout (light mode)

- **Header:** Lumen logo, Checker / Batch results, Light / Dark / System
- **Background:** Soft grey gradient + grid pattern
- **Card:** White panel with teal gradient strip on top
- **Home CTA:** Orange Check accessibility button
- **Batch CTA:** Teal Run a live scan button

Batch snapshot (2026-07-30)

Metric	Value
Websites tested	28
Passed	8 (4 clean · 4 with issues)
Failed	20
Total issues	374

Pass rule: Score ≥ 85 and Critical = 0.

Passed sites: Google UK, BBC iPlayer, BBC News, Disney+ UK, GitHub (balisikh), Google Maps, Wikipedia, example.com

MVP features

Area	Status
Scan UI, Playwright + axe	Done
Rate limiting, persistence, JSON export	Done
Optional AI tips, anonymous use	Done
PDF export / accounts / crawl	Not yet

Stack

- Next.js (React) + TypeScript
- Playwright (Chrome) + axe-core
- Postgres or PGLite
- Optional OpenAI-compatible AI tips

Generated 2026-08-03 from docs/screenshots and project README.